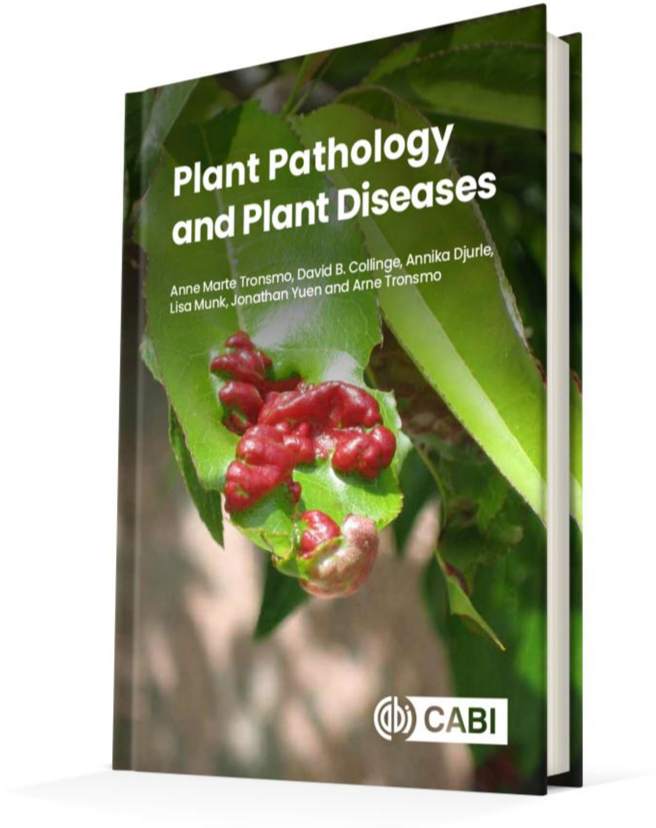

Plant Pathology and Plant Diseases

TEACHING MATERIALS

Fig. 11.1 Schematic model for of active defence. (© D.B.Collinge.)

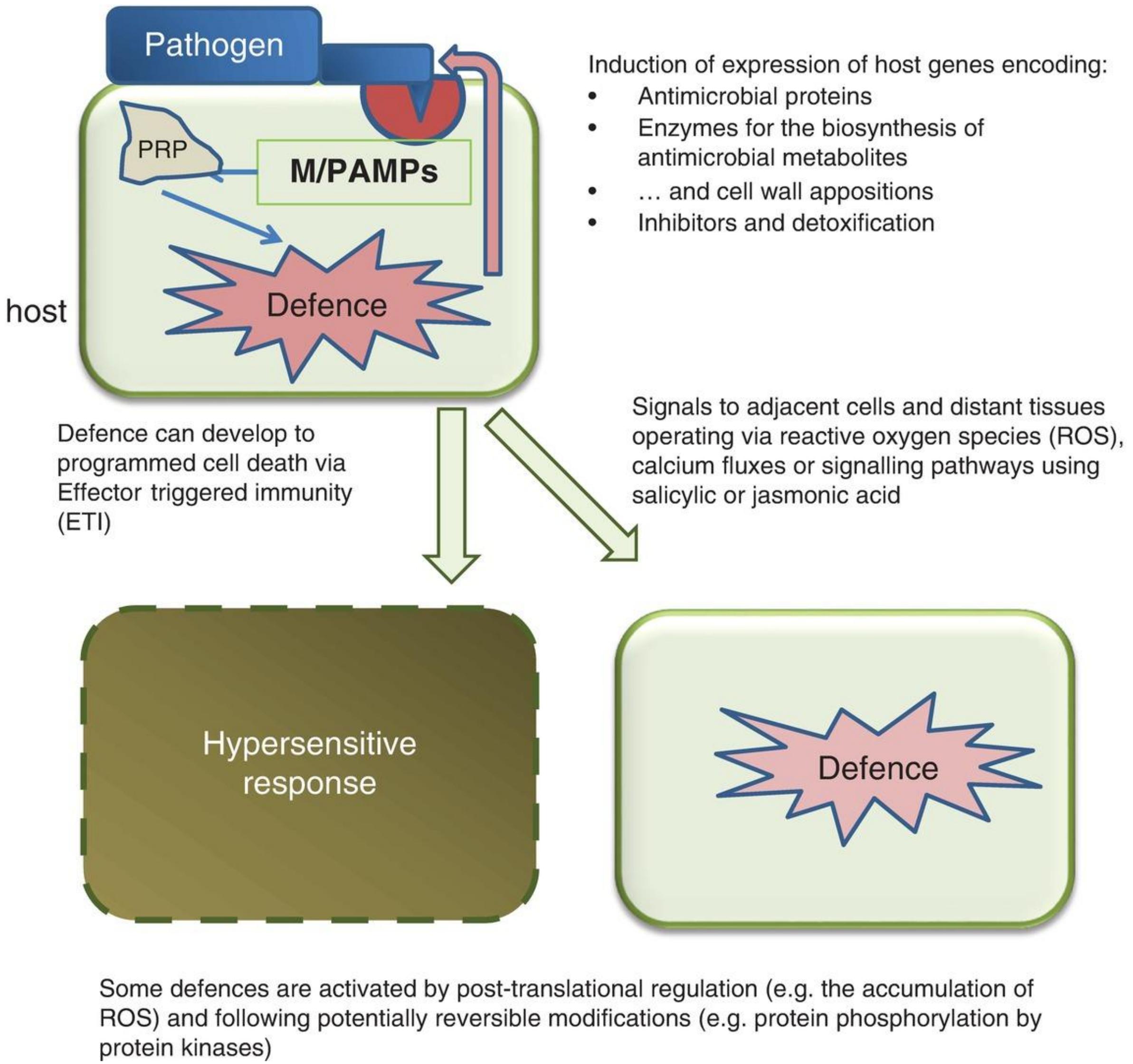

Fig. 11.2.A-D. Papillae (cell wall appositions) with different histochemical stains.
Epidermis of barley (left) and wheat (right) both inoculated with the barley
powdery mildew fungus, *Blumeria graminis* f.sp. *hordei* (24 hours after inoculation). (A) and (B) stained with lacmoid, indicating callose.
(From Wei *et al.*, 1994. *PMPP* 45. 467-484.)

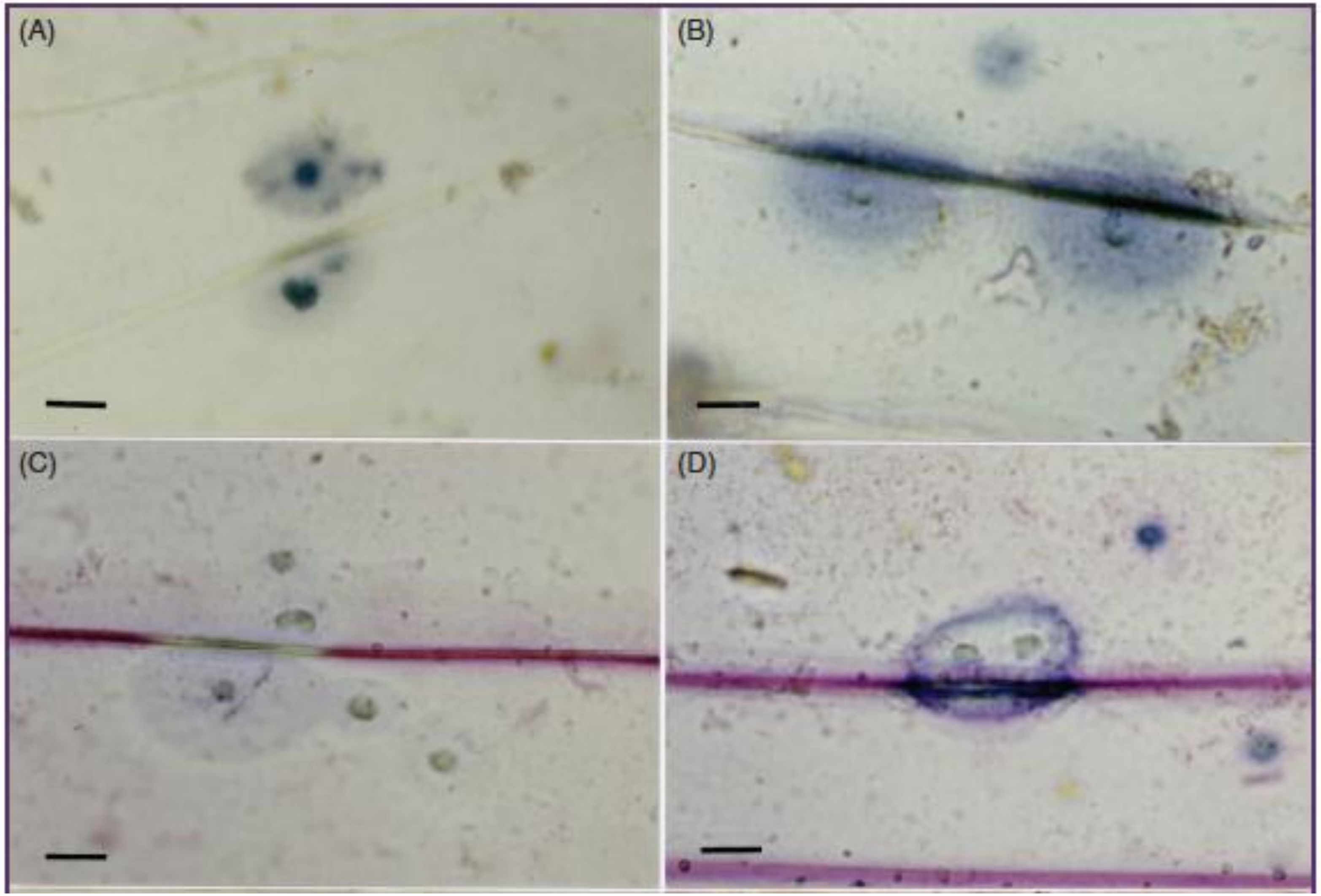

Fig. 11.2. E-H. Papillae (cell wall appositions) with different histochemical stains. Epidermis of barley (left) and wheat (right) both inoculated with the barley powdery mildew fungus, *Blumeria graminis* f.sp. *hordei* (24 hours after inoculation). E and F Phloroglucinol-HCl, indicating lignin.
G and H. Sakaguchi staining method, indicating guanidine compounds. Bar= 20 µm.
(From Wei *et al.*, 1994. *PMPP* 45. 467-484.)

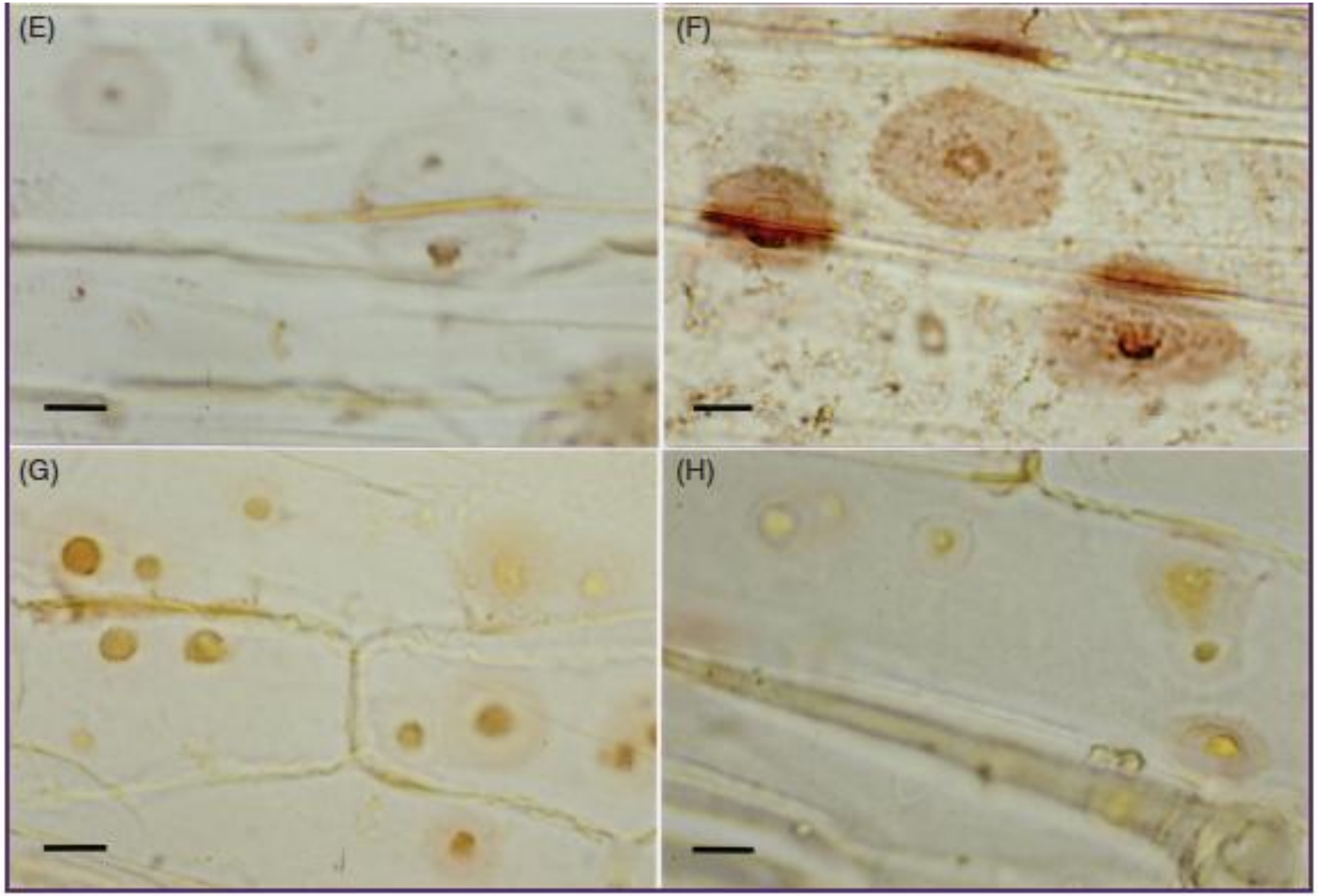

Fig. 11.3. DAB staining for papillae and the hypersensitive response (HR). Detection of reactive oxygen species using hydrogen peroxide in barley inoculated with the barley powdery mildew fungus, *Blumeria graminis* f.sp. *hordei* in papillae (A) 6 and (B) 24 hours after inoculation and (C) in cells exhibiting an HR 24 hours after inoculation. Diaminobenzidine (1 mg ml^{-1}) staining for 8 hours. P, primary germ tube; a, appresorial germ tube; pa, papilla; HR, hypersensitive response.
(From Thordal-Christensen *et al.*, 1997. *TPJ* 11. 1187-1194)

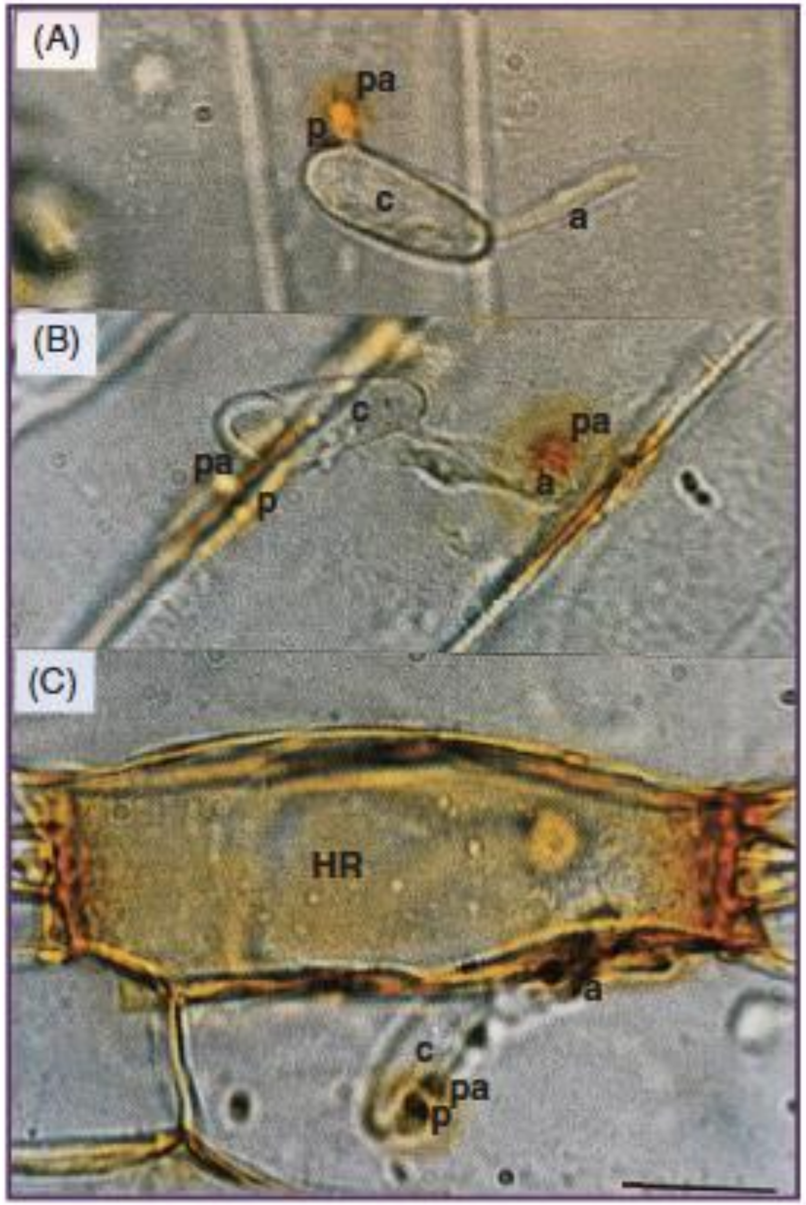

Fig. 11.4 Some structures of phytoalexins and phytoanticipins. (© D.B. Collinge.)